BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 112 · Merge Sort

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What does insertion sort do with the next unsorted item?

Q2. What is ROM commonly used to store?
A. An unsaved document
B. The current mouse position
C. A running game's score
D. Startup instructions

Q3. “Cache is a remote Internet server that replaces CPU registers.” Correct this. Explain why.

Q4. A pupil restarts a computer to open saved work. “RAM keeps every file forever.” Correct this. Why is secondary storage needed?

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I Do · Read new knowledge

Key vocabulary

Merge sort — Divide-and-conquer sort that splits and then merges sorted sublists.

Split — Divide a list into smaller parts.

Merge — Combine sorted sublists while preserving order.

Singleton — Sublist containing one element.

New learning

Divide and conquer: Merge sort repeatedly splits a list into smaller halves until each sublist has one item.

Single items are sorted: A one-element list is already in order by itself.

Merge in order: Take the smaller front item from two sorted sublists; copy it into the merged list.

Complete the sort: Repeat merging until there is one fully sorted list; do not confuse splitting with swapping.

Worked example

Show the major splitting and merging stages of merge sort for [8,3,5,1,7,2,6,4]. [6 marks]

1. Split into halves

2. [8,3,5,1] and [7,2,6,4], then split until single items remain.

3. Merge sorted pairs: [3,8], [1,5], [2,7], [4,6].

4. Merge quarters: [1,3,5,8], [2,4,6,7], then final [1,2,3,4,5,6,7,8]. [6]

We Do · Practise with guidance

Guided 1

Split [6,3,8,1] into singletons and merge into ascending order. [5]

Worked method: [6,3] and [8,1] [1]; [6],[3],[8],[1] [1]; [3,6],[1,8] [1]; [1,3,6,8] [2].

Guided 2

Merge two sorted sublists [2,7,9] and [1,3,10]. [4]

Worked method: Choose 1,2,3,7,9,10 [3]; resulting [1,2,3,7,9,10] [1].

Guided 3

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Explain why each half must be sorted before it can be merged in the usual merge step. [3]

Worked method: The merge chooses the smallest front item [1]; this is safe only when each sublist is ordered [1], producing a sorted merged list [1].

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You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Code First · Predict → Write → Test → Improve

Select which first element should be taken from two sorted lists.

```
array left = [4,10]
array right = [2,8]
if ... then
    print(left[0])
else
    print(right[0])
endif
```

write

Question 1

Split [7,2,5,1] to singletons and write the two sorted pairs. [4] Source: Original OCR J277-style practice question.

Question 2

Merge the already sorted lists [1,4,9] and [2,3,8]. State the resulting list. [3] Source: Original OCR J277-style practice question.

Question 3

Explain what divide-and-conquer means in merge sort. [3] Source: Original OCR J277-style practice question.

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Exit Ticket · Answer before checking

Binary search can safely discard half only when data is sorted.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Place the next item in the correct position in the sorted part. Shift larger items right.

Q2. D — Startup instructions

Q3. Cache is in or near the CPU. It gives faster access to often-used data and instructions than main memory.

Q4. Secondary storage keeps files and programs without power. Save the work there before restarting.

I Do / We Do pages · Guided practice

Q1. [6,3] and [8,1] [1]; [6],[3],[8],[1] [1]; [3,6],[1,8] [1]; [1,3,6,8] [2].

Q2. Choose 1,2,3,7,9,10 [3]; resulting [1,2,3,7,9,10] [1].

Q3. The merge chooses the smallest front item [1]; this is safe only when each sublist is ordered [1], producing a sorted merged list [1].

You Do pages · Code First

```
array left = [4,10]
array right = [2,8]
if left[0] <= right[0] then
    print(left[0])
else
    print(right[0])
endif
```

Expected result: 2

You Do pages · Independent questions

Q1. Split [7,2] and [5,1] [1]; singletons [7],[2],[5],[1] [1]; sorted pairs [2,7] [1] and [1,5] [1].

Q2. [1,2,3,4,8,9] [3; 1 for first three, 1 for last three, 1 for completely correct].

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Q3. Divide the list into smaller subproblems [1], solve them by sorting/merging sublists [1], combine to obtain one ordered result [1].

Exit Ticket · Model answer

True

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: